Tab 4: Risk Budgeting (Practice)

FitchLearning

Asset Class vs. Risk Class

Risk Budgeting with Asset Class and Risk Class Approaches
Hakan Kaya, Wai Lee and Yi Wan
The Journal of Investing Spring 2012, 21 (1) 109-115;

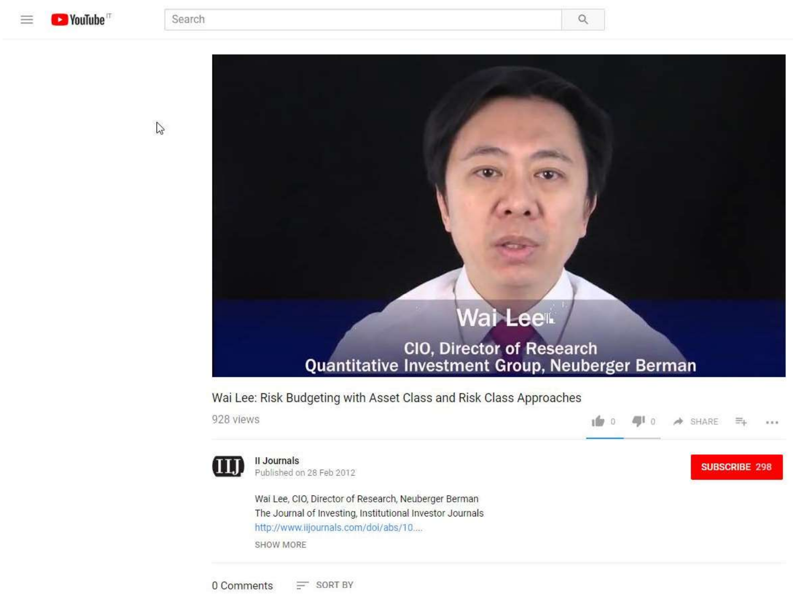
Image source: http://joi.iijournals.com/sites/default/files/styles/medium/public/highwire/iijinvest/21/1.cover-source.gif?itok=nM9XDU9v

Asset Class vs. Risk Class

“One of the natural responses to the most recent financial crisis is to point out the failure of modern portfolio theory and mean-variance optimization to provide diversification when it was most needed. Some critics, including pensions, foundations, and endowments, have proposed displacing “asset classes” with “risk classes” for the purpose of asset allocation. With this new risk class approach, investors determine an optimal mix of assets by achieving target exposures to different risks. From a conceptual standpoint, the risk class approach is superior to an asset class approach, as it recognizes that investable and tradable assets in a portfolio are merely vehicles for investors to gain exposures to a set of risks that are believed to be rewarded. However, practical application of the risk class approach presents its own set of challenges. If we consider that when the risk class approach is fully executed so that all risks are specified and idiosyncratic elements are almost negligible and uncorrelated, the approach converges with the asset class approach.”

Hakan Kaya, Wai Lee and Yi Wan. *Risk Budgeting with Asset Class and Risk Class Approaches*
The Journal of Investing Spring 2012, 21 (1) 109-115;

Asset Class vs. Risk Class



https://www.youtube.com/watch?v=bb2AYU0IALA

Appendix:
Code Samples

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Example 1:

```
% Modern Portfolio Theory
% Roncalli
% page 7
% Octave Manual p 635
% 25.2 Quadratic Programming

close all
clear all

% INPUTS
% repeat for phi = infinity, 5,2,1,0.5,0.20
phi=0.2;
gamma=1/phi;

mu=[5 6 8 6]'./100;
vol=[15 20 25 30]./100;
C=[1 0.1 0.4 0.5;0.1 1 0.7 0.4;0.4 0.7 1 0.8;0.5 0.4 0.8 1];
%compute covariance matrix
% Sigma=corr2cov(vol,C);
for i=1:4
  for j=1:4
    Sigma(i,j)=C(i,j)*vol(i)*vol(j);
  end
end

% Quadratic Programming
x0=[0 0 0 0]';
H=Sigma;
q=-gamma*mu;
A=ones(1,4);
b=ones(1,1);

[x, obj, info, lambda] = qp (x0, H, q, A, b)

%compute portfolio expected return
mu_p=x'*mu

%compute portfolio variance
var_p=x'*Sigma*x
vol_p=sqrt(var_p)
```

Example 2:

```
%modern portfolio theory
% Roncalli
% page 7
% Octave Manual p 635
% 25.2 Quadratic Programming

close all
clear all

% INPUTS
phi=1.61;
gamma=1/phi;

mu=[5 6 8 6]'./100;
vol=[15 20 25 30]./100;
C=[1 0.1 0.4 0.5;0.1 1 0.7 0.4;0.4 0.7 1 0.8;0.5 0.4 0.8 1];
%compute covariance matrix
% Sigma=corr2cov(vol,C);
for i=1:4
  for j=1:4
    Sigma(i,j)=C(i,j)*vol(i)*vol(j);
  end
end

% Quadratic Programming

x0=[0 0 0 0]';
H=Sigma;
q=-gamma*mu;
A=ones(1,4);
b=ones(1,1);
lb=[0 0 0 0]';
ub=[10 10 10 10]';

[x, obj, info, lambda] = qp (x0, H, q, A, b,lb,ub)


%compute portfolio expected return
mu_p=x'*mu

%compute portfolio variance
var_p=x'*Sigma*x
vol_p=sqrt(var_p)
```

Example 3:

```
%modern portfolio theory
% Roncalli
% page 9
% Octave Manual p 635
% 25.2 Quadratic Programming

close all
clear all

% INPUTS
phi=1.97;
gamma=1/phi;

mu=[5 6 8 6]'./100;
vol=[15 20 25 30]./100;
C=[1 0.1 0.4 0.5;0.1 1 0.7 0.4;0.4 0.7 1 0.8;0.5 0.4 0.8 1];
%compute covariance matrix
% Sigma=corr2cov(vol,C);
for i=1:4
  for j=1:4
    Sigma(i,j)=C(i,j)*vol(i)*vol(j);
  end
end

% Quadratic Programming

x0=[0 0 0 0]';
H=Sigma;
q=-gamma*mu;
A=ones(1,4);
b=ones(1,1);
lb=[0 0 0 0]';
ub=[0.4 0.4 0.4 0.4]';

[x, obj, info, lambda] = qp (x0, H, q, A, b,lb,ub)

%compute portfolio expected return
mu_p=x'*mu

%compute portfolio variance
var_p=x'*Sigma*x
vol_p=sqrt(var_p)
```

Example 4:

```
% Modern portfolio theory
% Roncalli
% page 11
% Octave Manual p 635
% 25.2 Quadratic Programming

% DeFusco et al page 448
% three asset example

close all
clear all

phi=100;
N=3;
x=[0.32 0.51 0.17]';
mu=[16 16.1 17.6]'./100;
vol=[29 24.7 27.3]./100;
C=[1 0.361 0.317;0.361 1 0.548 ;0.317 0.548 1];

%compute covariance matrix
for i=1:N
  for j=1:N
    Sigma(i,j)=C(i,j)*vol(i)*vol(j);
  end
end

x_star=(inv(Sigma)*ones(N,1))/(ones(1,N)*inv(Sigma)*ones(N,1))
+1/phi*((ones(1,N)*inv(Sigma)*ones(N,1))*inv(Sigma)*mu-
(ones(1,N)*inv(Sigma)*ones(N,1))*inv(Sigma)*ones(N,1))/(ones(
1,N)*inv(Sigma)*ones(N,1))

x_mv=(inv(Sigma)*ones(N,1))/(ones(1,N)*inv(Sigma)*ones(N,1))

mu_pi=x_mv'*mu
var_pi=x_mv'*Sigma*x_mv
vol_pi=sqrt(var_pi)
```

Example 5:

```
% VAR and ES
% Roncalli
% page 75

close all
clear all

% three stcosk A,B,C
p=[244 135 315]';
x=[0.5203 0.1439 0.3358]';
mu=[50 30 20]'./10000;
vol=[2 3 1]./100;
rho=[1 0.5 0.25;0.5 1 0.6;0.25 0.6 1];
for i=1:3
  for j=1:3
    Sigma(i,j)=rho(i,j)*vol(i)*vol(j);
  end
end


alpha=0.99;

disp('in percent')

% compute VAR
VAR_x = -x'*mu + norminv(alpha)*sqrt(x'*Sigma*x)

% compute ES
ES_x = -x'*mu + (sqrt(x'*Sigma*x))/(1-alpha)*normpdf(norminv(alpha))


disp('in dollars')
% two stocks of A, one of B and one of C
p=[244*2 135*1 315*1]';

% compute VAR
VAR_p = -p'*mu + norminv(alpha)*sqrt(p'*Sigma*p)

% compute ES
ES_p = -p'*mu + (sqrt(p'*Sigma*p))/(1-alpha)*normpdf(norminv(alpha))
```

Example 6:

```
% Risk Contribution (RC)
% Roncalli
% page 80

close all
clear all

% three assets A,B,C

% p=[244 135 315]';

x=[0.50 0.20 0.30]';
mu=[0 0 0]'./10000;
vol=[30 20 15]./100;
rho=[1 0.8 0.5;0.8 1 0.3;0.5 0.3 1];

for i=1:3
  for j=1:3
    Sigma(i,j)=rho(i,j)*vol(i)*vol(j);
  end
end

%compute portfolio variance
var_p=x'*Sigma*x
vol_p=sqrt(x'*Sigma*x)

% RISK CONTRIBUTIONS VOLATILITY
%compute marginal volatilities
disp('stdev')
MR = (Sigma*x)/sqrt(x'*Sigma*x)

%compute risk contributions
RCi = x.*(Sigma*x)/sqrt(x'*Sigma*x)

%verify sum RC equals volatility
RC = sum(RCi)

alpha=0.99;

% RISK CONTRIBUTIONS VAR
disp('VAR')
RCi_VAR = x.*(-mu+norminv(alpha)*(Sigma*x)/sqrt(x'*Sigma*x))
RC_VAR = sum(RCi_VAR)


% RISK CONTRIBUTIONS ES
disp('ES')
RCi_ES =  x.*(-mu+(Sigma*x)/(1-
alpha)/sqrt(x'*Sigma*x)*normpdf(norminv(alpha)))
RC_ES = sum(RCi_ES)
```

Example 7:

```
% Risk Contribution (RC)
% Roncalli
% page 102

close all
clear all

% three assets A,B,C
x=[391926 273737 685779]';
alpha=0.99;


% page 102 Introduction to Risk Parity and Budgeting
% 2.2.1.2 Solving the non-linear system of risk budgeting contraints

function obj = phi(x)

mu=[10 5 9]'./100;
vol=[30 20 15]./100;
rho=[1 0.8 0.5;0.8 1 0.3;0.5 0.3 1];
B=[250000 100000 150000]';
for i=1:3
  for j=1:3
    Sigma(i,j)=rho(i,j)*vol(i)*vol(j);
  end
end
alpha=0.99;
b=B./sum(B);
obj = sum((x.*(-mu+(Sigma*x)/(1-alpha)/sqrt(x'*Sigma*x)*normpdf(norminv(alpha)))-b.*( -x'*mu +
(sqrt(x'*Sigma*x))/(1-alpha)*normpdf(norminv(alpha)))).^2);

endfunction

function r = g (x)
r = sum(x)-1;
endfunction

% Nonlinear Programming Octave manuale page 637
x0=[0.2 0.2 0.2]';
lb=[0.01 0.01 0.01];
ub=[1 1 1];
[x, obj, info, lambda] = sqp (x0, @phi,@g,[],lb,ub)



 disp('____ES weights_____')

mu=[10 5 9]'./100;
vol=[30 20 15]./100;
rho=[1 0.8 0.5;0.8 1 0.3;0.5 0.3 1];
B=[250000 100000 150000]';
for i=1:3
  for j=1:3
    Sigma(i,j)=rho(i,j)*vol(i)*vol(j);
  end
end
 % RISK CONTRIBUTIONS ES
RCi_ES =  x.*(-mu+(Sigma*x)/(1-alpha)/sqrt(x'*Sigma*x)*normpdf(norminv(alpha)))
RC_ES = sum(RCi_ES)
x
sum(x)


 disp('____ES dollars_____')
x=[391926 273737 685779]';
% RISK CONTRIBUTIONS ES
RCi_ES =  x.*(-mu+(Sigma*x)/(1-alpha)/sqrt(x'*Sigma*x)*normpdf(norminv(alpha)))
RC_ES = sum(RCi_ES)
x
sum(x)
```

Example 8:

```
% Risk Contribution (RC)
% Roncalli
% RB Table 2.20

close all
clear all

% two assets A,B


x=[0.60 0.40]';
% mu=[0 0 0]'./10000;
vol=[0.2 0.3];
vol_ew=0.2179;
vol_erc=0.2078;
vol_mv=0.1964;

rho=[1 0.5;0.5 1];

for i=1:2
  for j=1:2
    Sigma(i,j)=rho(i,j)*vol(i)*vol(j);
  end
end


%compute portfolio variance
var_p=x'*Sigma*x
vol_p=sqrt(x'*Sigma*x)


% RISK CONTRIBUTIONS
disp('EW')
x=[0.5 0.5]'
MR = (Sigma*x)/vol_ew
%compute risk contributions
RCi = x.*(Sigma*x)/vol_ew
%verify sum RC equals volatility

% RISK CONTRIBUTIONS
disp('ERC')
x=[0.6 0.4]'
MR = (Sigma*x)/vol_erc
%compute risk contributions
RCi = x.*(Sigma*x)/vol_erc
%verify sum RC equals volatility

% RISK CONTRIBUTIONS
disp('MV')
x=[0.8571 0.1429]'
MR = (Sigma*x)/vol_mv
%compute risk contributions
RCi = x.*(Sigma*x)/vol_mv
%verify sum RC equals volatility
```

Example 9:

```
% Risk Contribution (RC)
% Braga
% equations 3.20
% RB Table 3.3

close all
clear all

% two assets A,B
% vol=[0.0402 0.1247];
volA=0.0402;
volB=0.1247;

wA=(1/volA)/(1/volA+1/volB)
wB=(1/volB)/(1/volA+1/volB)

rho=[1 -0.12;-0.12 1];
vol=[volA volB];

for i=1:2
  for j=1:2
    Sigma(i,j)=rho(i,j)*vol(i)*vol(j);
  end
end

x=[wA wB]';

%compute portfolio variance
var_p=x'*Sigma*x
vol_p=sqrt(x'*Sigma*x)


% RISK CONTRIBUTIONS
disp('ERC or Navie RP')
MR = (Sigma*x)/sqrt(x'*Sigma*x)
%compute risk contributions
RCi = x.*(Sigma*x)/sqrt(x'*Sigma*x)
%verify sum RC equals volatility
```